Ex.No.-1.1-Design and Draft the given 2D Sketches in modelling software.

AIM:

To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1.Autodesk fusion 360
2.Windows 11
3.16 GB of RAM (integrated graphics recommend 2 GB or more)
4.2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1.Open a new file and then select the sketch icon.
2.Select the plane for sketching
3.Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch

Center Circles & Central Features:

● Central Star / Multi-lobe Profile: Inner lobes have a radius of R6.
● Outer Diameter Arc: Diametral reference of Ø40 and Ø66 around the central body.

Side Lobe / Circular Projections (Left & Right):

● Center-to-Center Distance (Horizontal): 47 mm on both the left side and right side from the central vertical axis (total span = 94 mm).

Vertical Offsets from Centerline:

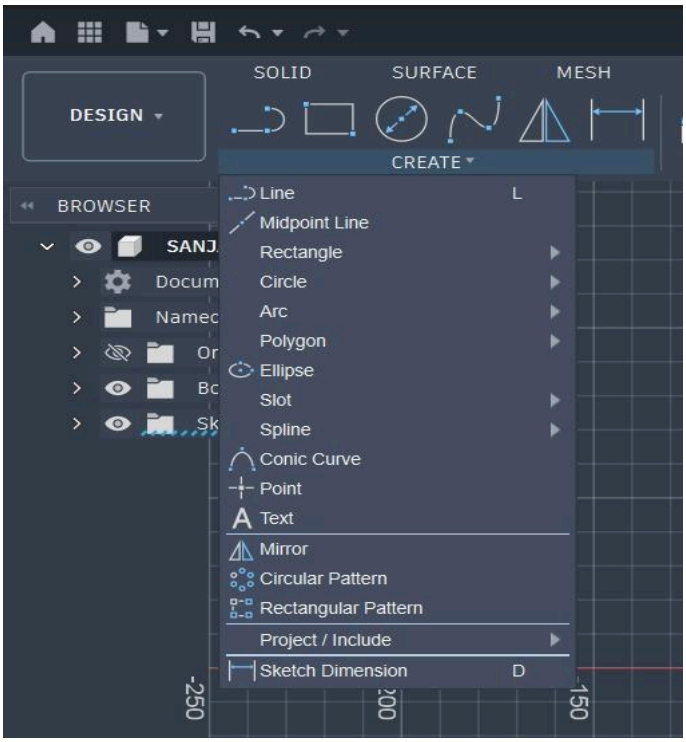
○ 26 mm upward vertical distance to the upper reference.
○ 24 mm downward vertical distance to the lower feature.

Left/Right Outer Loops:

○ Outer circular arcs of radius R7.
○ Inner holes with a diameter of Ø10.
○ Connecting neck fillets/radii of R10.

Bottom & Transitional Curves:

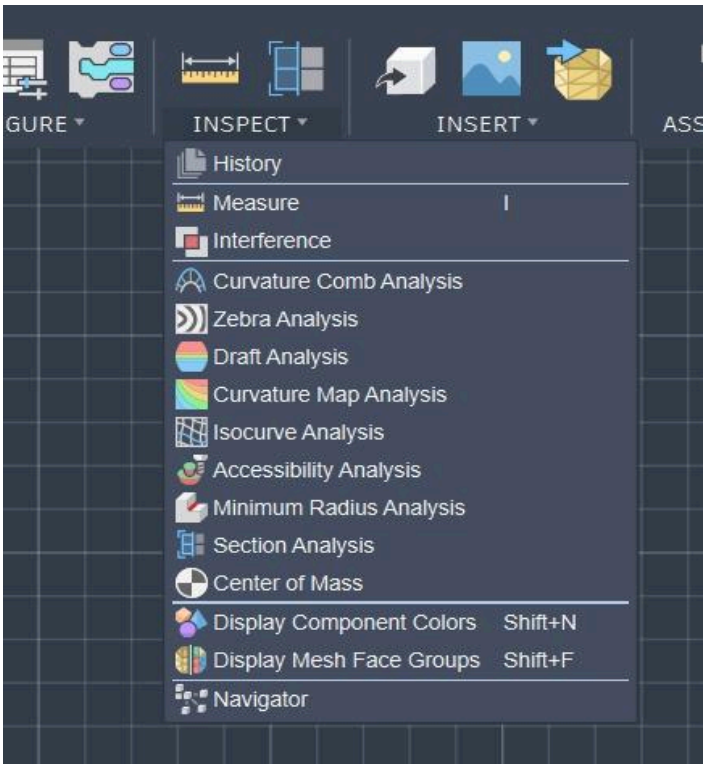
● Bottom central hole diameter of Ø10.
● Surrounding lower hub/flange diameter of Ø20.
● Transitional connecting curves/fillets with radii of R20.
● Upper-left curved profile transition of radius R20



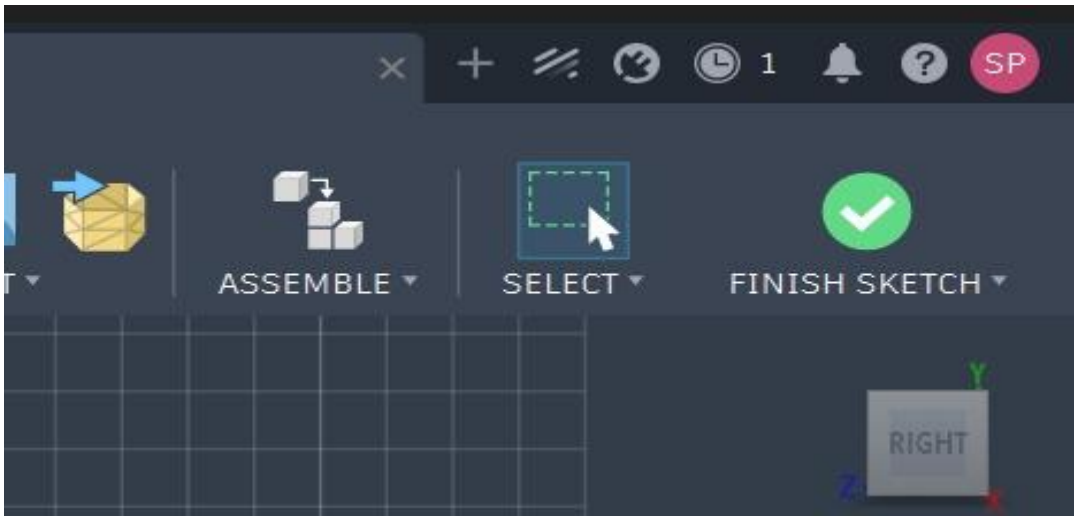
4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch.



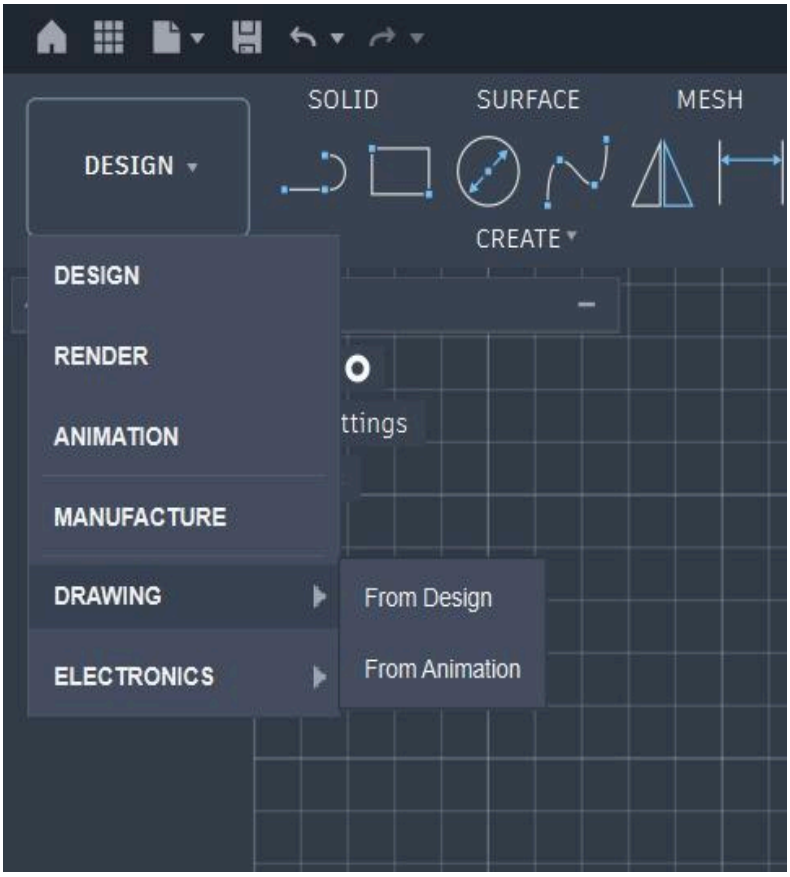
5.Use inspect feature to measure the dimensions of the sketch.

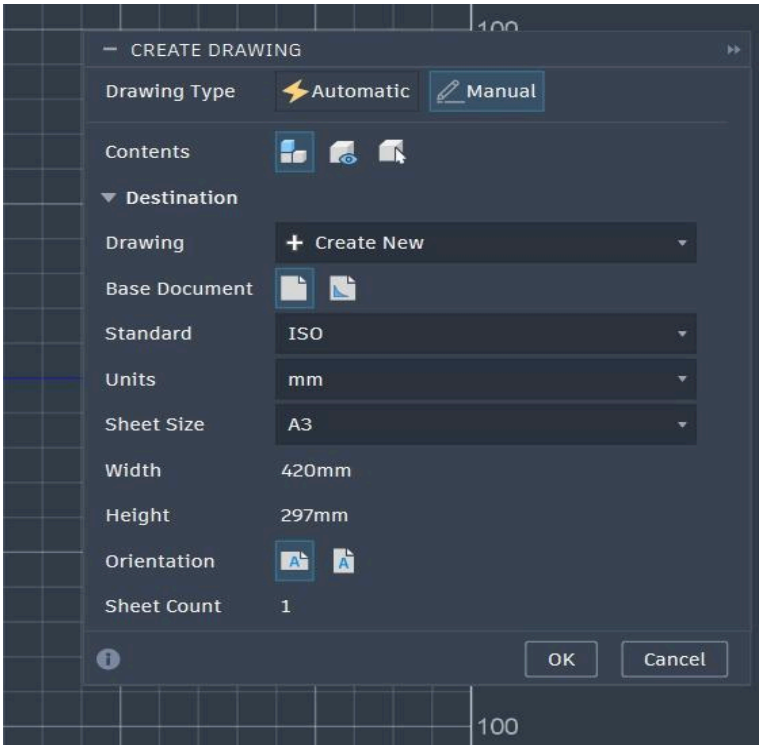


6.After finishing the sketch use the finish sketch option to finalize the sketch.

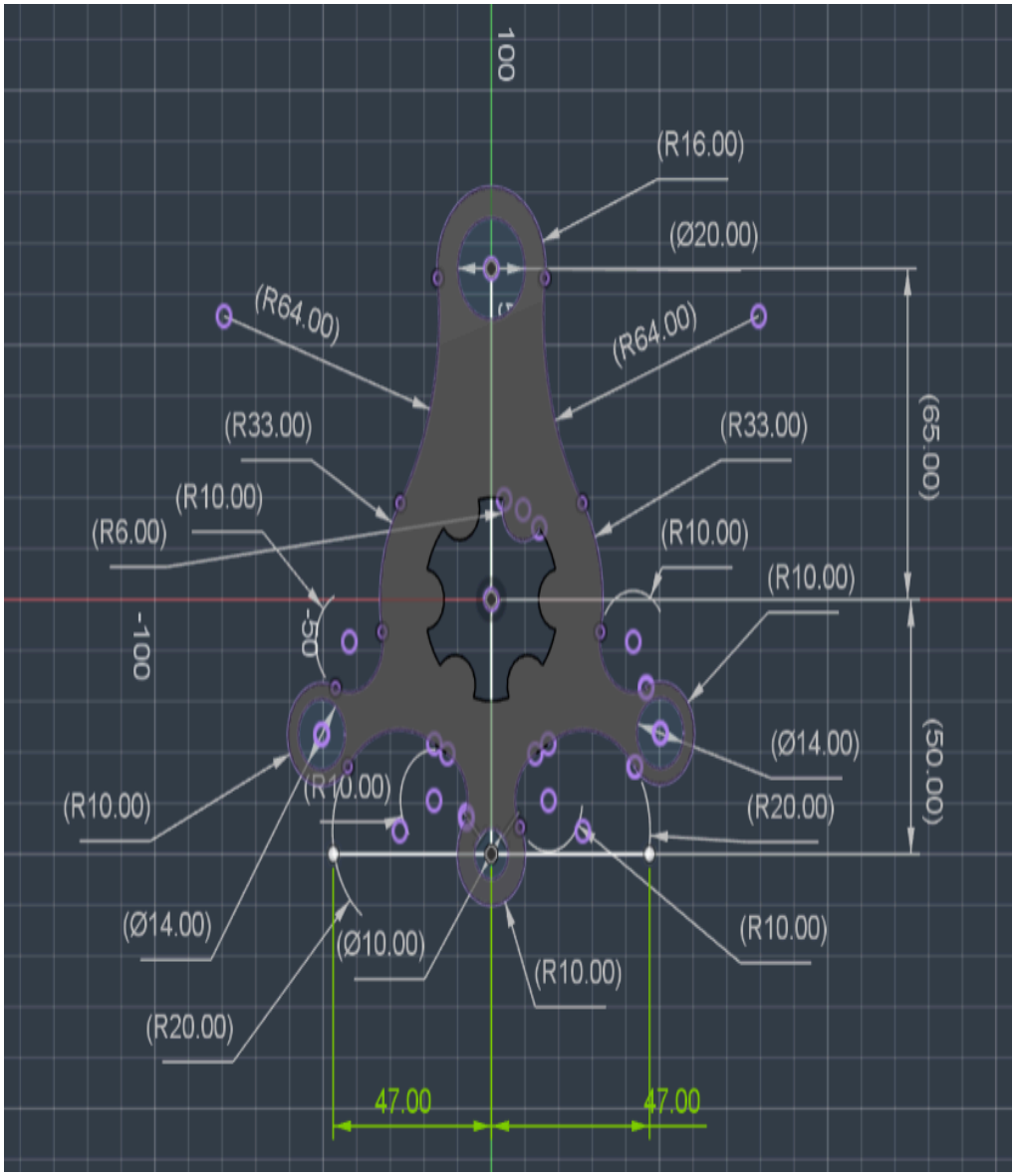


7.After finishing the sketch, switch to drawing mode to plot the sketch in ISO standard.

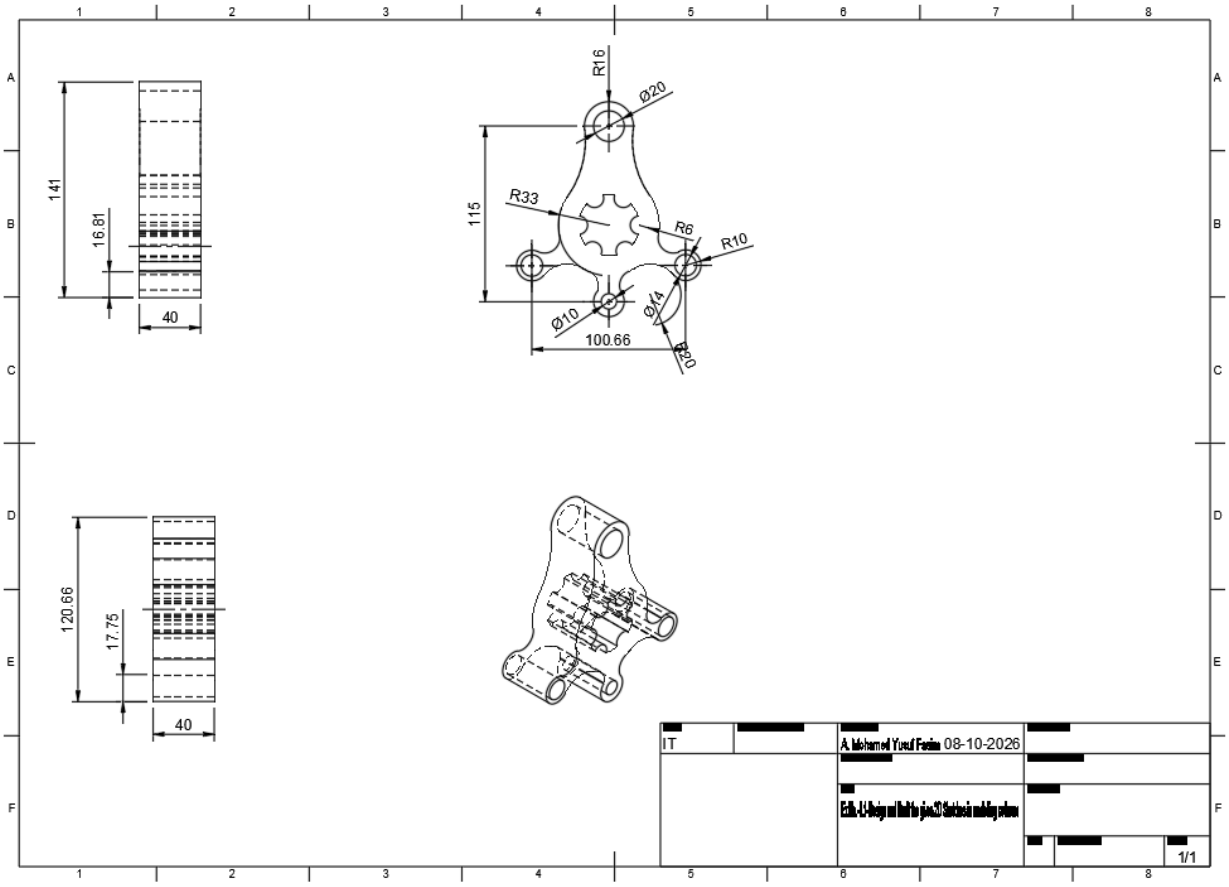

SOLID
SURFACE
MESH
DESIGN
CREATE
DESIGN
RENDER
ANIMATION
MANUFACTURE
DRAWING
ELECTRONICS
From Design
From Animation


CREATE DRAWING
Drawing Type
Automatic
Manual
Contents
Destination
Drawing
Create New
Base Document
Standard
ISO
Units
mm
Sheet Size
A3
Width
420mm
Height
297mm
Orientation
Sheet Count
1
OK
Cancel

Sketches:


100
(R16.00)
(Ø20.00)
(R64.00)
(R64.00)
(R33.00)
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(R10.00)
(R10.00)
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47.00
47.00

OUTPUT:



RESULT:

Thus the given sketch is drawn and drafted using fusion 360 tool.